

Moaaz Ibrahim Reda

FullStack Developer

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PROFESSIONAL SUMMARY

Detail-oriented Computer Science student (GPA: 3.8) specializing in Backend Development with ASP.NET Core, C#, Docker, and SQL Server. Completed intensive training programs with DEPI and ITI, gaining hands-on expertise in building RESTful Web APIs, containerizing backend applications, implementing clean software architecture, and applying Machine Learning models.

EDUCATION

Bachelor of Computer and Information Science, University of Ain Shams

09/2024 – Present

Expected Graduation: July 2028 | Cumulative GPA: 3.8 / 4.0

Cairo, Egypt

PRACTICAL EXPERIENCE & TRAINING

Digital Egypt Pioneers Initiative (DEPI) – Round 5

07/2026 – 12/2026

Backend Development Trainee (.NET & Cloud Technologies)

6 Months

- Mastered backend software development utilizing SQL Server, C#, ASP.NET Core Web API, Docker containerization, and Git/GitHub version control workflows.

Information Technology Institute (ITI)

06/2025 – 07/2025

Frontend Web Development Trainee

1 Month

- Gained hands-on web development experience building dynamic, responsive UI applications using React, JavaScript (ES6+), HTML5, and CSS3.

SKILLS

Programming Languages:

C#, C++, SQL, JavaScript, TypeScript, Python, HTML5, CSS3

Backend & Web Technologies:

ASP.NET Core Web API, Entity Framework Core (EF Core), SQL Server, React 19, TailwindCSS

AI & Data Science:

Scikit-Learn, Pandas, NumPy, Machine Learning, K-Nearest Neighbors (KNN), Random Forest

Architecture & Design:

OOP, SOLID Principles, Design Patterns (GoF), Clean Code, RESTful APIs

DevOps & Tools:

Docker, Git, GitHub, VS Code, Postman, Codeforces

Soft Skills & Languages:

Problem Solving, Team Collaboration | Arabic (Native), English (Intermediate)

PROJECTS

Interactive World Atlas (Team Project)

03/2026 – 05/2026

Full-Stack Web App | ASP.NET Core Web API | EF Core | SQL Server | Google Gemini AI | React 19 | TypeScript

- Built an interactive world map featuring real-time country detection using GeoJSON ray casting and Web Mercator projection.
- Integrated Google Gemini AI to automatically generate and structure country data from Wikipedia content, persisting entities via EF Core.
- Automated media pipeline with custom DuckDuckGo image scraper and Wikipedia media extractor.
- Implemented URL-based routing with automatic map zoom to country bounding boxes on direct navigation.

Census Income Prediction (Team AI Project)

11/2025 – 12/2025

Machine Learning Project | Python | Scikit-Learn | Pandas | NumPy | KNN | Random Forest

- Engineered an end-to-end ML classification pipeline in Python predicting income levels (> \$50K) from US Census data.
- Optimized K-Nearest Neighbors (KNN) model achieving 85.27% accuracy via Grid Search with 5-Fold Cross-Validation (Manhattan metric, K=31).
- Implemented data preprocessing with median/mode imputation, IQR outlier clipping, standard scaling, and feature engineering (net capital, marital status mapping, safe label encoding).
- Evaluated feature importances against Random Forest (86.71% accuracy) to analyze non-linear census attribute relationships.

Metro Management System (Team Project)

03/2025 – 07/2025

Data Structures Project | C++ | Graph Algorithms & Queue Data Structures

- Designed and implemented a metro transit system in C++ for operational scheduling using Graphs (BFS/Dijkstra) and Queues.
- Structured clean object-oriented architecture to reduce memory overhead and optimize execution performance.

Banking System (Team Project)

10/2025 – 11/2025

OOP Project | C++ | Object-Oriented Design & Security

- Engineered a C++ banking application with transaction management, deposit/withdrawal validation, and balance tracking.
- Enforced Role-Based Access Control (RBAC) to restrict administrative permissions and secure sensitive operations.

ACTIVITIES & VOLUNTEER

Open Source Community (OSC) – Ain Shams University

11/2024 – Present

Science and Technology Member

Cairo, Egypt

- Applied software architecture principles (SOLID, DRY, KISS, YAGNI) and GoF Design Patterns to build scalable software solutions.

ACM Student Chapter – Ain Shams University

02/2024 – 02/2025

Competitive Programming Member

Cairo, Egypt

- Trained in structured problem-solving on Codeforces divisions to strengthen algorithmic thinking and implementation speed.